

Unification and Resolution

Ex No: 7

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AIM

To study and implement unification and resolution exercises using Prolog facts, compound terms, queries, and rules.

Exercise 1: Sentence Matcher

% Facts

sentence(the_cat_sits).

sentence(the_dog_runs).

% Query

% ?- sentence(X).

Output:

X = the_cat_sits ; X = the_dog_runs ; false.

Exercise 2: Compound Term Unification

% Facts

likes(john, food(pizza, spicy)).

likes(mary, food(pizza, sweet)).

% Query

% ?- likes(john, food(pizza, T)).

Output:

T = spicy.

Exercise 3: Conflict Resolution

% Facts

rule1(climate_change).

rule2(climate_change).

% Rule

conflict(X) :- rule1(X), rule2(X).

% Query

% ?- conflict(climate_change).

Output:

True.

RESULT

Thus, the unification and resolution exercises were implemented successfully and the expected Prolog outputs were obtained.

